

PAVAN RAJ R

AI/ML Engineer | Computer Vision Specialist | NLP & Automation

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with hands-on industry experience at Bosch in computer vision and industrial automation. Skilled in building end-to-end intelligent systems including real-time image processing pipelines, NLP-based voice assistants, deep learning models, and test automation frameworks. Proficient in Python, OpenCV, TensorFlow, PyTorch, and API-driven development. Passionate about deploying scalable ML solutions that solve real-world problems and improve operational efficiency.

WORK EXPERIENCE

Intern – AI/ML & Computer Vision | Bosch

Bangalore, India | Jan 2026 – May 2026

- Developed a Glow Plug Inspection System using Masked Template Matching for automated detection of components in industrial production trays.
- Built real-time image processing pipelines using OpenCV and Python, integrated with Pypylon SDK for industrial camera capture.
- Implemented Non-Maximum Suppression (NMS) to enhance detection accuracy and reduce false positives in production environments.
- Designed a Tkinter-based GUI supporting live inspection, ROI selection, and template management for operator use.
- Developed a CSV-based structured logging system for tracking inspection outcomes (PASS/FAIL) with timestamp and traceability.
- Reduced manual inspection errors significantly through automation of quality control workflows on the production floor.

PROJECTS

1. Glow Plug Inspection System – Industrial Automation

Tech Stack: Python, OpenCV, NumPy, Tkinter, Pypylon SDK

- Designed an end-to-end automated inspection system for detecting glow plugs in industrial production trays.
- Applied masked template matching combined with ROI filtering to significantly improve detection precision over standard methods.
- Implemented multi-template support with rotation handling (0° and 180°) to accommodate various component orientations.
- Engineered structured detection with left-to-right sorting and tray validation logic for consistent, ordered output.
- Delivered a full pipeline covering image capture, processing, pass/fail decision-making, and result logging.

2. AI Voice Assistant using NLP, Speech Recognition & Google Gemini API

Tech Stack: Python, NLP, SpeechRecognition, PyQt5, Gemini API, Multithreading

- Designed and developed a real-time AI voice assistant with continuous listening and intelligent command execution.
- Implemented NLP pipelines and integrated Gemini API for context-aware, smart responses to user queries.
- Built a PyQt5 custom UI with a non-blocking multithreaded architecture for seamless user interaction.
- Added system automation capabilities: screenshots, window control, screen lock, and file operations.
- Applied modular design pattern separating GUI, audio processing, and AI logic for maintainability and scalability.

3. Real-Time Face Recognition & Deep Learning–Based Emotion Classification

Tech Stack: Python, OpenCV, DeepFace, LBPH, NumPy

- Built a real-time computer vision system using live webcam feed for face recognition and emotion classification.
- Implemented LBPH algorithm for identity detection and DeepFace for multi-class emotion classification (happy, sad, angry, neutral).
- Enabled dynamic face registration without retraining the full dataset, allowing rapid enrollment of new users.
- Developed dedicated training (train.py) and inference (main.py) modules for clean pipeline separation.

4. Web Automation Testing Framework using Selenium WebDriver & Python

Tech Stack: Selenium, Python, PyCharm

- Built automation test scripts for Orange HRM (open-source HR platform), covering login, dashboard interaction, and form validations.
- Used Selenium WebDriver for DOM interactions with Python scripting for robust and maintainable test coverage.
- Designed reusable test functions to improve maintainability, reduce redundancy, and accelerate debugging cycles.

TECHNICAL SKILLS

Programming Languages: Python, Java, OOP

AI & Machine Learning: Scikit-Learn, TensorFlow, PyTorch, Deep Learning, Feature Engineering, Model Training, Classification, Regression

Computer Vision & NLP: OpenCV, DeepFace, LBPH, Masked Template Matching, NMS, SpeechRecognition, NLP Pipelines

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, PyQt5, Tkinter, Pypylon SDK

Tools & Platforms: Git, GitHub, PyCharm, Jupyter Notebook, Selenium WebDriver

Databases: MySQL, SQL Queries

APIs & Integrations: Google Gemini API, Open Trivia DB API, REST APIs

EDUCATION

Master of Computer Applications (MCA) | S-VYASA Deemed University Bangalore | 2024 – 2026
(Pursuing)

Specialization: Artificial Intelligence, Machine Learning & Data Science

Bachelor of Science (BSc) | Surana Autonomous College

Bangalore | 2022 – 2024

Major: Physics & Computer Science | CGPA: 7.88

SOFT SKILLS

Problem Solving • Critical Thinking • Teamwork & Collaboration • Adaptability • Communication